

PART 1: AI CONCEPTS

Q1) What is Artificial Intelligence?

A) Artificial Intelligence means making computers and machines work like humans. It helps machines learn, think, understand, and solve problems on their own.

Q2) Difference between AI, Machine Learning, Generative AI, and Agentic AI

A) Artificial Intelligence (AI)

-> AI is a broad technology that makes machines act smart like humans.

Ex: Voice assistants, face recognition

Machine Learning (ML)

-> ML is a part of AI where machines learn from data and improve automatically.

Ex: YouTube recommendation

Generative AI

-> Generative AI creates new content like text, images, music, or videos.

Ex: ChatGPT writing answers or AI image generators creating pictures

Agentic AI

-> Agentic AI can make decisions and perform tasks on its own to achieve a goal.

Ex: AI booking tickets or managing schedules automatically

Q3) Explain any three open-source AI models available on Hugging Face.

A) The three open- source AI models available on Hugging face are: Llam , Gemma , Deepseek

Q4) What is the difference between traditional programming and AI-assisted development

A) Traditional Programming:

- > Humans write all the code manually
- > Computer follows fixed instructions
- > Takes more time for development
- > Debugging is mostly manual
- > Requires strong coding knowledge
- > Works only on programmed rules

AI-Assisted Development

- > AI helps in writing code
- > AI can suggest and generate code automatically
- > Development becomes faster
- > AI helps find and fix errors
- > Easier for beginners and developers
- > AI learns patterns from data and improves suggestions

Q5) Explain AI hallucinations with one example

A) AI hallucination happens when an AI system gives false or incorrect information as if it were true. The AI creates answers that sound confident but are actually wrong.

Ex:



